

KINDERSLEY, SK, Canada

WMO: 711290

Lat: 51.520N

Lon: 109.180W

Elev: 2276

StdP: 13.53

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-20.5	-33.6	0.9	-26.0	-28.1	1.2	-20.4	30.2	19.3	26.6	15.1	6.6	190	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	87.3	63.6	83.8	62.7	80.3	61.5	67.3	80.5	65.3	78.1	63.5	76.0	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	93.5	72.3	60.8	86.4	69.9	58.8	80.6	68.0	33.1	80.9	31.5	78.3	30.0	75.9	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.3	23.2	20.6	DB	-32.6	93.9	5.2	3.3	-36.3	96.3	-39.4	98.2	-42.3	100.1	-46.1	102.4	(4)
(5)			WB	-32.8	71.1	5.1	2.7	-36.5	73.1	-39.5	74.7	-42.4	76.2	-46.1	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	8.0	10.3	23.1	39.4	51.6	59.7	65.1	63.6	54.3	40.0	23.8	11.3		
	DBStd	23.37	15.39	13.90	14.55	10.02	8.26	6.33	5.65	6.48	8.60	9.44	12.40	14.32		
	HDD50	6034	1302	1110	833	334	82	5	0	1	50	330	787	1200		
	HDD65	10149	1767	1530	1298	767	421	181	69	104	334	776	1237	1665		
	CDD50	1534	0	0	0	17	130	297	468	424	179	19	0	0		
	CDD65	172	0	0	0	0	4	22	71	62	13	0	0	0		
	CDH74	2641	0	0	0	10	150	332	912	926	303	8	0	0		
	CDH80	803	0	0	0	1	28	79	282	318	95	0	0	0		
	WSAvg	9.8	9.2	9.1	10.6	11.3	11.2	10.3	8.9	8.8	9.5	10.0	9.6	9.1		
	PrecAvg	13.9	0.4	0.3	0.5	0.9	1.4	3.2	2.5	2.0	1.1	0.6	0.5	0.4		
	PrecMax	25.0	1.4	0.7	1.2	2.4	3.3	5.4	7.2	5.2	4.3	1.9	1.3	1.2		
(15) Precipitation	PrecMin	5.4	0.0	0.0	0.0	0.1	0.0	0.9	0.7	0.0	0.2	0.2	0.0	0.0		
	PrecStd	3.8	0.3	0.2	0.3	0.6	0.9	1.2	1.4	1.1	0.9	0.4	0.4	0.3		
	0.4% DB	40.4	39.2	58.0	74.5	84.0	87.2	91.7	91.9	88.5	74.0	56.5	43.5			
	0.4% MCWB	36.1	35.1	46.1	52.6	57.6	62.6	67.1	63.9	61.6	54.7	45.6	37.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	36.9	36.1	50.8	67.6	78.2	81.7	86.8	87.5	82.3	65.8	48.7	38.5			
	2% MCWB	33.7	33.1	41.9	49.9	56.0	61.0	65.7	62.6	59.0	50.7	41.1	34.5			
	5% DB	33.6	33.2	44.9	62.0	73.5	77.7	82.9	83.4	76.6	60.0	43.2	35.1			
	5% MCWB	31.1	30.9	38.7	46.3	54.3	59.5	64.5	62.7	57.0	47.9	37.4	32.0			
	10% DB	29.4	29.8	39.9	56.8	68.9	73.7	79.1	79.1	71.2	55.0	38.9	30.4			
	10% MCWB	27.5	28.0	35.7	43.9	51.8	57.8	63.2	61.3	54.6	45.4	34.9	28.3			
	0.4% WB	36.6	35.7	47.4	54.1	60.0	66.5	71.1	68.0	63.6	56.1	46.6	38.1			
	0.4% MCDB	40.0	38.5	56.6	71.0	78.4	80.8	84.1	82.5	83.6	71.1	55.1	42.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.9	33.3	42.4	50.6	57.4	63.7	68.5	66.0	60.2	51.7	41.6	34.7			
	2% MCDB	36.7	35.7	49.8	65.8	75.3	76.0	81.6	80.5	78.0	62.9	47.9	38.2			
	5% WB	31.2	31.0	39.0	47.6	55.2	61.5	66.5	64.2	57.7	48.6	38.0	31.9			
	5% MCDB	33.5	33.2	44.4	59.9	70.5	73.6	78.6	78.1	74.0	58.6	42.5	35.1			
	10% WB	27.5	28.0	35.8	44.9	53.0	59.5	64.6	62.3	55.5	45.9	35.1	28.3			
	10% MCDB	29.4	29.8	39.5	55.2	66.7	70.4	76.1	75.8	69.7	54.3	38.5	30.3			
(35) Mean Daily Temperature Range	MDBR	17.0	17.9	17.5	22.3	24.1	21.1	23.2	24.5	24.4	21.4	16.7	16.5			
	5% DB	19.0	18.7	23.0	31.7	31.4	27.7	29.3	32.3	34.1	30.1	22.5	19.9			
	5% WB	16.8	16.4	15.5	16.9	14.8	12.2	12.7	13.2	15.3	17.1	15.7	16.5			
	MCDBR	19.2	18.6	22.2	29.1	27.5	23.5	25.5	26.4	31.2	28.4	20.8	19.9			
	MCWBR	17.2	16.5	15.4	16.3	13.5	11.5	12.6	12.1	14.6	16.8	15.1	16.8			
(40) Clear-Sky Solar Irradiance	taub	0.261	0.269	0.299	0.358	0.382	0.381	0.379	0.393	0.340	0.293	0.265	0.247			
	taud	2.307	2.356	2.410	2.268	2.247	2.309	2.345	2.262	2.392	2.505	2.455	2.364			
	Ebn at Noon	242	276	287	279	276	277	276	264	268	265	244	232			
	Edh at Noon	20	25	29	38	41	39	37	38	30	22	17	16			
(44) All-Sky Solar Radiation	RadAvg	370	704	1135	1532	1862	1917	2070	1672	1210	727	373	267			
	RadStd	43	60	92	132	130	148	94	135	111	95	40	31			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page